

WIELUN, Poland

WMO: 124550

Lat: 51.2102N

Lon: 18.5566E

Elev: 659

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	12.4	1.6	6.1	8.3	6.6	7.9	14.1	22.2	40.0	19.8	38.8	5.3	90	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.2	67.5	82.7	66.0	79.7	64.7	69.5	81.6	68.0	79.0	66.5	76.6	6.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	97.0	73.8	64.2	92.1	72.5	62.7	87.4	70.7	33.9	81.9	32.7	79.1	31.5	76.8	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.2	13.4	DB	1.1	92.5	6.9	2.8	-3.9	94.5	-7.9	96.2	-11.8	97.8	-16.7	99.8	(4)
(5)			WB	0.5	73.0	6.7	2.1	-4.3	74.5	-8.3	75.7	-12.0	76.9	-16.9	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	29.5	32.0	38.5	48.7	57.0	62.7	66.5	65.9	57.8	48.9	40.2	32.6	(6)
(7)		DBStd	15.05	9.36	8.95	7.93	7.85	7.04	6.36	5.95	5.66	6.34	7.68	7.54	8.45	(7)
(8)		HDD50	2597	634	504	361	115	17	0	0	0	8	116	301	541	(8)
(9)		HDD65	6299	1099	924	820	490	261	117	53	58	229	498	744	1006	(9)
(10)		CDD50	2035	0	0	6	76	235	381	512	494	241	83	7	0	(10)
(11)		CDD65	262	0	0	0	1	14	48	100	87	12	0	0	0	(11)
(12)		CDH74	2413	0	0	0	28	157	465	928	727	107	1	0	0	(12)
(13)		CDH80	633	0	0	0	2	19	120	285	188	19	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.8	7.5	7.6	6.8	5.9	5.6	5.5	5.3	5.7	6.4	7.1	7.8	(14)
(15)	Precipitation	PrecAvg	23.8	1.4	1.3	1.5	1.5	2.8	2.9	3.6	2.1	2.0	1.5	1.7	1.5	(15)
(16)		PrecMax	33.1	3.1	2.3	3.8	3.2	8.4	5.9	12.4	4.9	5.3	3.5	3.4	4.7	(16)
(17)		PrecMin	15.4	0.3	0.2	0.6	0.0	1.1	0.9	1.3	0.6	0.7	0.2	0.0	0.4	(17)
(18)		PrecStd	4.3	0.6	0.6	0.8	0.7	1.5	1.3	2.4	1.0	1.1	0.9	0.7	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	53.7	64.1	77.9	82.5	88.9	91.6	89.7	82.8	72.6	60.0	52.4	(19)
(20)			MCWB	46.0	47.1	51.1	59.9	63.9	69.7	68.6	68.3	63.6	60.4	53.4	48.1	(20)
(21)		2%	DB	46.5	49.9	58.4	71.4	78.4	83.3	86.8	84.8	76.9	67.9	56.6	48.6	(21)
(22)			MCWB	43.3	45.2	48.6	56.2	62.6	67.4	67.5	67.0	62.3	58.8	51.4	45.6	(22)
(23)		5%	DB	43.7	46.6	53.9	66.4	74.6	79.3	83.0	81.4	73.0	63.7	53.4	45.5	(23)
(24)			MCWB	40.9	42.9	46.6	53.2	61.0	65.1	66.3	65.8	60.7	56.7	49.5	43.1	(24)
(25)		10%	DB	41.2	43.7	50.0	62.3	70.7	75.6	79.4	77.9	69.3	59.9	50.5	42.8	(25)
(26)			MCWB	39.0	40.4	44.5	51.4	58.9	63.0	65.0	64.3	59.4	54.7	47.5	40.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	48.0	52.5	60.9	66.7	71.5	71.5	70.8	65.7	62.5	54.4	48.9	(27)
(28)			MCDB	49.7	50.9	61.0	74.1	78.2	84.3	84.8	82.8	76.4	69.3	58.5	51.7	(28)
(29)		2%	WB	43.6	45.7	49.6	57.6	64.2	68.8	69.7	68.8	63.9	59.5	52.2	45.8	(29)
(30)			MCDB	46.2	49.4	56.5	69.1	74.5	80.7	81.6	80.7	73.5	65.7	56.2	48.2	(30)
(31)		5%	WB	41.2	43.2	47.2	54.9	62.2	66.7	68.1	67.4	62.2	57.3	49.8	43.3	(31)
(32)			MCDB	43.5	46.5	52.9	64.1	72.2	76.7	78.8	78.3	71.0	62.8	52.8	45.4	(32)
(33)		10%	WB	39.0	40.5	45.0	52.6	60.2	64.5	66.6	65.8	60.3	55.1	47.6	40.9	(33)
(34)			MCDB	41.0	43.3	49.7	60.5	69.1	73.5	76.5	75.2	67.3	59.3	50.2	42.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	9.7	12.7	17.0	17.7	17.9	18.2	15.7	12.9	9.1	7.6	(35)	
(36)			MCDBR	9.2	13.1	19.5	24.3	24.0	24.2	23.7	21.9	18.3	13.4	9.4	(36)	
(37)			MCWBR	7.7	9.5	11.9	12.0	11.4	11.1	9.0	9.5	9.9	10.5	9.4	7.5	(37)
(38)		5% WB	MCDBR	9.1	12.6	17.5	21.7	21.4	21.8	20.7	20.8	19.0	16.9	12.7	9.1	(38)
(39)			MCWBR	7.8	9.7	11.5	11.2	11.0	11.1	8.8	9.2	9.5	10.3	9.3	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.343	0.381	0.431	0.424	0.420	0.446	0.434	0.399	0.371	0.347	0.312	(40)	
(41)		taud	2.281	2.271	2.216	2.135	2.207	2.251	2.191	2.228	2.295	2.337	2.338	2.324	(41)	
(42)		Ebn at Noon	217	244	255	255	264	266	256	252	247	231	204	200	(42)	
(43)		Edh at Noon	22	29	36	44	43	42	43	40	33	27	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	242	468	845	1308	1622	1702	1602	1435	1029	590	276	186	(44)	
(45)		RadStd	46	83	108	172	178	167	194	120	143	89	36	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.90	N/A	N/A	+1.33	N/A	N/A	N/A	-293	+133	+49

Nomenclature: See separate page